

Year 7 · Chapter 9 · The number plane: answers (teacher copy)

9.01 Grid references on maps (booklet p. 3)

WE DO examples (teacher)

- Ex 1a: the lake
- Ex 1b: the library
- Ex 2a: D9
- Ex 2b: B2, C2, B3, C3
- Ex 3a: e.g. walk to Gauss Rd, west to Euclid St, then north
- Ex 3b: it is a convention so everyone reads it the same way

Easy (drill)

- | | | |
|----------|-----------|--------|
| 1. A5 | 11. Park | 21. B2 |
| 2. D1 | 12. Bus | 22. C4 |
| 3. E4 | 13. Post | 23. H2 |
| 4. G5 | 14. Oval | 24. H6 |
| 5. F6 | 15. Wharf | 25. A1 |
| 6. H6 | 16. Farm | 26. E2 |
| 7. Zoo | 17. Pool | 27. B6 |
| 8. Gym | 18. Fire | 28. F6 |
| 9. Bank | 19. G3 | 29. H5 |
| 10. Cafe | 20. E5 | 30. F1 |

Medium (drill)

- B5
- E6
- H3
- E5 (Club)
- F5
- G4 (Oval)
- F3
- E5 (Club)
- C5
- G3 (Cafe)

Challenging

- own map
- own directions

Extension

- C5
- 18 squares

Exit ticket

- Level 1: stadium
- Level 2: F2
- Level 3: student centre (H7)

9.02 Points in the first quadrant (booklet p. 9)

WE DO examples (teacher)

- Ex 1a: plotted
- Ex 1b: plotted on the axes
- Ex 2a: P(4, 1), Q(2, 5)
- Ex 2b: No: (2, 5) is 2 across, 5 up
- Ex 3a: rectangle
- Ex 3b: (0, 0)

Easy (drill)

- | | | |
|-------|------------|------------|
| 1. 6 | 8. 7 | 15. (4, 6) |
| 2. 10 | 9. 8 | 16. (4, 5) |
| 3. 4 | 10. 4 | 17. (2, 6) |
| 4. 1 | 11. 10 | 18. (6, 6) |
| 5. 12 | 12. 0 | 19. (1, 8) |
| 6. 12 | 13. (7, 8) | 20. (6, 0) |
| 7. 6 | 14. (0, 0) | 21. (8, 8) |

22. (2, 8)

23. (1, 2)

24. (0, 3)

Medium (drill)

- | | | |
|------|------|-------|
| 1. B | 5. C | 9. K |
| 2. J | 6. E | 10. L |
| 3. M | 7. D | 11. A |
| 4. F | 8. H | 12. G |

Challenging

- (1, 6)
- x-axis points have $y = 0$; y-axis points have $x = 0$

Extension

- (3, 5)
- 24 square units

Exit ticket

- Level 1: 7
- Level 2: on the y-axis, 5 up
- Level 3: (1, 5)

9.03 The number plane (booklet p. 15)

WE DO examples (teacher)

- Ex 1a: 1st top right, 2nd top left, 3rd bottom left, 4th bottom right
- Ex 1b: 3rd
- Ex 2a: A(2, 3), B(-3, 1), C(-2, -4), D(4, -2)
- Ex 2b: 4th
- Ex 3a: a zigzag line
- Ex 3b: e.g. (-2, 3); e.g. (0, 5)

Easy (drill)

- | | | |
|--------|--------------|--------------|
| 1. 3rd | 9. 2nd | 17. (-1, -1) |
| 2. 2nd | 10. 4th | 18. (5, -2) |
| 3. 2nd | 11. 2nd | 19. (-1, -2) |
| 4. 4th | 12. 3rd | 20. (3, -1) |
| 5. 4th | 13. (4, 3) | 21. (-1, -3) |
| 6. 3rd | 14. (3, -4) | 22. (4, 4) |
| 7. 3rd | 15. (-5, -1) | 23. (0, -2) |
| 8. 2nd | 16. (-1, -5) | 24. (-4, 3) |

Medium (drill)

- | | | |
|-------------|-------------------|-------------------|
| 1. (7, -7) | 11. (8, 7) | 21. 4th |
| 2. (3, 3) | 12. (5, 6) | 22. 4th |
| 3. (-4, 2) | 13. (6, -8) | 23. 2nd |
| 4. (-4, 1) | 14. (-2, -4) | 24. on the x-axis |
| 5. (-1, 4) | 15. (-1, 8) | 25. 3rd |
| 6. (4, 6) | 16. (-5, 2) | 26. 2nd |
| 7. (-7, -3) | 17. 1st | 27. 3rd |
| 8. (-4, 4) | 18. on the y-axis | 28. the origin |
| 9. (4, -8) | 19. 1st | |
| 10. (2, -6) | 20. 2nd | |

Challenging

- B: 2nd
- (0, 0)
- (-3, -2)

Extension

- 12 square units
- (1, -1), (4, -1), (1, -3)

Exit ticket

- Level 1: 2nd
- Level 2: e.g. (3, -1)
- Level 3: (-2, -5)

9.04 Plotting tables of values (booklet p. 21)

WE DO examples (teacher)

1. Ex 1a: vertical line $x = -2$
2. Ex 1b: horizontal line $y = 3$
3. Ex 2a: straight line going down
4. Ex 2b: straight line going up steeply
5. Ex 3a: y goes up by 2
6. Ex 3b: $y = 7$

Easy (drill)

- | | | |
|-------------------------------------|----------------------------|----------------------------|
| 1. (0, 5), (1, 9), (2, 13), (3, 17) | 5. (6, 25)
6. (4, 16) | 13. (6, 15)
14. (4, 12) |
| 2. (0, 0), (1, 3), (2, 6), (3, 9) | 7. (4, 4)
8. (6, 30) | 15. (3, 14)
16. (6, 9) |
| 3. (0, 5), (1, 7), (2, 9), (3, 11) | 9. (3, 16)
10. (3, 5) | 17. (3, 10)
18. (4, 15) |
| 4. (0, 2), (1, 3), (2, 4), (3, 5) | 11. (3, 15)
12. (4, 17) | 19. (6, 28) |

Medium (drill)

- | | | |
|-------|--------|--------|
| 1. 24 | 6. 28 | 11. 9 |
| 2. 13 | 7. 3 | 12. 19 |
| 3. 15 | 8. 24 | 13. 8 |
| 4. 19 | 9. 25 | 14. 13 |
| 5. 29 | 10. 10 | 15. 8 |

Challenging

1. a U-shaped curve (parabola)
2. no, it is a curve

Extension

1. -5
2. $y = 4x$

Exit ticket

1. Level 1: (0, 3), (1, 5), (2, 7)
2. Level 2: horizontal line
3. Level 3: it goes up by 3

9.05 Tables of values from rules (booklet p.

27)

WE DO examples (teacher)

1. Ex 1a: 2, 3, 4, 5, 6
2. Ex 1b: -11, -9, -7, -5, -3
3. Ex 2a: 6, 5, 4, 3, 2
4. Ex 2b: -5, -2, 1, 4, 7
5. Ex 3a: $y = 2x + 4$
6. Ex 3b: $y = 8 - x$

Easy (drill)

- | | | |
|-------|-------------------|-------------------|
| 1. 15 | 8. 5 | 15. 6, 7, 8, 9 |
| 2. 14 | 9. 19 | 16. 6, 10, 14, 18 |
| 3. 16 | 10. 8, 11, 14, 17 | 17. 0, 5, 10, 15 |
| 4. 6 | 11. 4, 8, 12, 16 | 18. 3, 5, 7, 9 |
| 5. 22 | 12. 8, 9, 10, 11 | 19. 9, 10, 11, 12 |
| 6. 2 | 13. 9, 12, 15, 18 | |
| 7. 5 | 14. 3, 7, 11, 15 | |

Medium (drill)

- | | | |
|------------|-------------|--------------|
| 1. $x = 2$ | 6. $x = 8$ | 11. $x = 6$ |
| 2. $x = 9$ | 7. $x = 4$ | 12. $x = 7$ |
| 3. $x = 8$ | 8. $x = 1$ | 13. $x = 10$ |
| 4. $x = 1$ | 9. $x = 0$ | 14. $x = 0$ |
| 5. $x = 5$ | 10. $x = 6$ | 15. $x = 9$ |

Challenging

1. $y = 2x + 4$
2. $y = 8 - x$

Extension

1. 2, 0, 0, 2, 6
2. $y = x^2$

Exit ticket

1. Level 1: 3
2. Level 2: -1, 3, 7
3. Level 3: $y = 3x + 1$

9.06 Graphing linear equations (booklet p. 33)

WE DO examples (teacher)

1. Ex 1a: $y = -3, -1, 1, 3$; straight line
2. Ex 1b: line through (0, 3) going down
3. Ex 2a: yes
4. Ex 2b: yes
5. Ex 3a: $y = -4, -3, -2, -1, 0$
6. Ex 3b: (0, -1)

Easy (drill)

- | | | |
|--------|---------------------------------------|--------------------------------------|
| 1. yes | 9. yes | 14. (-1, -2), (0, 1), (1, 4), (2, 7) |
| 2. yes | 10. (-1, -3), (0, -1), (1, 1), (2, 3) | 15. (-1, -1), (0, 0), (1, 1), (2, 2) |
| 3. yes | 11. (-1, -4), (0, -1), (1, 2), (2, 5) | 16. (-1, -3), (0, 0), (1, 3), (2, 6) |
| 4. no | 12. (-1, 2), (0, 3), (1, 4), (2, 5) | 17. (-1, -1), (0, 2), (1, 5), (2, 8) |
| 5. yes | 13. (-1, 0), (0, 2), (1, 4), (2, 6) | |
| 6. yes | | |
| 7. yes | | |
| 8. no | | |

Medium (drill)

- | | | |
|-------|---------|--------|
| 1. -4 | 7. 11 | 13. 9 |
| 2. 3 | 8. 2 | 14. 19 |
| 3. 0 | 9. -4 | 15. -1 |
| 4. 4 | 10. -10 | 16. 10 |
| 5. -6 | 11. -1 | |
| 6. 0 | 12. 12 | |

Challenging

1. parallel lines, 2 units apart
2. (1, 2)

Extension

1. $m = 3$
2. $y = 2x + 1$ and $y = 2x - 3$

Exit ticket

1. Level 1: 5
2. Level 2: yes
3. Level 3: yes